

Lawyers are Dishonest? Quantifying Representational Harms in Commonsense Knowledge Resources¹

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Abstract⁶

Warning: this paper contains content that may be offensive or upsetting.⁷

Commonsense knowledge bases (CSKB) are increasingly used for various natural language processing tasks. Since CSKBs are mostly human-generated and may reflect societal biases, it is important to ensure that such biases are not conflated with the notion of commonsense. Here we focus on two widely used CSKBs, ConceptNet and GenericsKB, and establish the presence of bias in the form of two types of representational harms, *overgeneralization* of polarized perceptions and representation *disparity* across different demographic groups in both CSKBs. Next, we find similar representational harms for downstream models that use ConceptNet. Finally, we propose a filtering-based approach for mitigating such harms, and observe that our filtered-based approach can reduce the issues in both resources and models but leads to a performance drop, leaving room for future work to build fairer and stronger commonsense models.⁸

1 Introduction⁹

Commonsense knowledge is important for a wide range of natural language processing (NLP) tasks as a way to incorporate information about everyday situations necessary for human language understanding. Numerous models have included knowledge resources such as ConceptNet (Speer et al., 2017) for question answering (Lin et al., 2019), sarcasm generation (Chakrabarty et al., 2020), and dialogue response generation (Zhou et al., 2018, 2021), among others. However, commonsense knowledge resources are mostly human-generated, either crowdsourced from the public (Speer et al., 2017; Sap et al., 2019) or crawled from massive web corpora (Bhaktavatsalam et al., 2020). For example, ConceptNet originated from the Open

Source	Examples
ConceptNet	(lady, UsedFor, fxxk)
	(lawyer, RelatedTo, dishonest) (church, UsedFor, brain washing)
GenericsKB	(Chinese people are very reclusive group of people) (Lawyers are registered menace to society)

Table 1: Biased cases in ConceptNet and GenericsKB.¹¹

Mind Common Sense project that collects commonsense statements online from web users (Singh et al., 2002)¹ and GenericsKB consists of crawled text from public websites. One issue with this approach is that the crowdsourcing workers and web page writers may conflate their own prejudices with the notion of commonsense. For instance, we have found that querying for some target words such as “church” as shown in Table 1 in ConceptNet, results in biased triples.¹³

The potentially biased nature of commonsense knowledge bases (CSKB), given their increasing popularity, raises the urgent need to quantify biases both in the knowledge resources and in the downstream models that use these resources. We present the first study on measuring bias in two large CSKBs, namely ConceptNet (Speer et al., 2017), the most widely used knowledge graph in commonsense reasoning tasks, and GenericsKB (Bhaktavatsalam et al., 2020), which expresses knowledge in the form of natural language sentences and has gained increasing usage. We formalize a new quantification of “representational harms,” *i.e.*, how social groups (referred to as “targets”) are perceived (Barocas et al., 2017; Blodgett et al., 2020) in the context of CSKBs.¹⁴

We consider two types of such harms in the context of CSKBs. One is *intra-target overgeneralization*, indicating that “common sense” in these resources may unfairly attribute a polarized (nega-

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¹ConceptNet also includes knowledge from expert-created sources such as WordNet (Miller, 1995)

tive or positive) characteristic to all members of a target class such as “*lawyers are dishonest*.” The other is *inter-target disparity*, occurring when targets have significantly different coverage in the CSKB in terms of both the number of statements about the targets (e.g., “*Persian*” might have much fewer CS statements than “*British*”) and perception toward the targets (“*islam*” might have more negative CS statements than “*christian*”).

We propose a quantification of *overgeneralization* and *disparity* in CSKBs using two proxy measures of polarized perceptions: sentiment and regard (Sheng et al., 2019). Applying the proposed metrics of bias to ConceptNet and GenericsKB, we find harmful overgeneralizations of both negative and positive perceptions over many target groups, indicating that human biases have been conflated with “*common sense*” in these resources. We find severe disparities across the targets in demographic categories such as professions and genders, both in the number of statements and the polarized perceptions about the targets.

We then examine two generative downstream tasks and the corresponding models that use ConceptNet. Specifically, we focus on automatic knowledge graph construction and story generation and quantify biases in COMeT (Bosselut et al., 2019) and CommonsenseStoryGen (CSG) (Guan et al., 2020). We find that these models also contain the harmful overgeneralizations and disparities found in ConceptNet. We then design a simple mitigation method that filters unwanted triples according to our measures in ConceptNet. We retrain COMeT using filtered ConceptNet and show that our proposed mitigation approach helps in reducing both overgeneralization and disparity issues in the COMeT model but leads to a performance drop in terms of the quality of triples generated according to human evaluations. We open-source our data and prompts to evaluate biases in commonsense resources and models for future work ².

2 Quantifying Representational Harms ⁴

2.1 Representational Harms ⁵

Representational harms occur “*when systems reinforce the subordination of some groups along the lines of identity*” and can be further categorized into stereotyping, recognition, denigration, and underrepresentation (Barocas et al., 2017; Crawford,

2017; Blodgett et al., 2020). This work aims to formalize representational harms specifically for a set of statements about some target groups (Nadeem et al., 2020), e.g. “*lawyer is related to dishonest*” is a statement about the group “*lawyer*.”

When measuring such harms, we consider the core concept of *polarized perceptions*: non-neutral views that can take the form of either prejudice that expresses negative views ³ or favoritism that expresses positive views toward a certain target perceived in the statement (Mehrabi et al., 2021).

More formally, let $\mathbb{S} = \{s_1, s_2, \dots, s_n\}$ indicate the set of n natural language statements s_i and let $\mathbb{T} = \{t_1, t_2, \dots, t_m\}$ indicate the set of m targets such that each t_j has appeared at least once in $\mathbb{S}$. Each statement s_i contains a target t_j . We use $s_i^{+/-}(t_j)$ to indicate that the statement expresses a positive or negative perception toward the target t_j . We are interested in quantifying the representational harms toward $\mathbb{T}$ in this set $\mathbb{S}$.

2.2 Two Types of Harms ¹⁰

To adapt the definition of representational harms to a sentence set, we define two sub-types of harms, *intra-target overgeneralization* and *inter-target disparity*, aiming to cover different categories of representational harms (Barocas et al., 2017; Crawford, 2017). We consider *overgeneralization* that directly examines whether targets such as “*lawyer*” or “*lady*” are perceived positively or negatively in the statements (examples in Table 1), covering categories including stereotyping, denigration, and favoritism. Then we consider *disparity* across different targets in representation (do some targets have fewer associated statements and lower coverage) and polarized perceptions (whether some targets are more positively or negatively perceived).

Intra-target Overgeneralization The ideal sentence set depicting a target group such as “*lawyer*” or “*lady*” should have neither *favoritism* nor *prejudice* toward the target group. Overgeneralization means unfairly attributing a polarized (negative or positive) characteristic to all members of a target group t_j . In the context of sentence sets $\mathbb{S}$, we define overgeneralization to be polarized sentences in the set regarding the targets. When the statement s_i contains negatively-polarized views toward a target group such as “*lawyers are dishonest*,” it

²https://github.com/Ninarehm/Commonsense_bias

³The word *prejudice* is defined as “*preconceived (usually unfavorable) evaluation of another person*” (Lindzey and Aronson, 1968). We focus on only unfavorable prejudice and use favoritism for favorable evaluation.

demonstrates *prejudice* toward lawyers. It is an overgeneralized statement because it implies that *all* lawyers are not honest. The same logic applies to positively-polarized statements such as “*British people are brilliant*” that constitute favoritism.

To measure the polarization in a sentence s_i , we use two approximations: sentiment polarity and regard (Sheng et al., 2019). We define an intra-target overgeneralization score using these measures for every target in $\mathbb{S}$. Formally, for each target t_j , we collect all statements in $\mathbb{S}$ that contain target t_j to form a target-specific set $\mathbb{S}_{t_j}$, we then apply the sentiment and regard classifiers individually on all statements in $\mathbb{S}_{t_j}$ and report the average percentages of statements that are classified as positive or negative sentiment or regard labels for the target t_j . The negative and positive overgeneralization bias (prejudice and favoritism) for t_j is quantified as:

$$O^-(\mathbb{S}, t_j) = |\mathbb{S}_{t_j}^-| |\mathbb{S}_{t_j}|^{-1} \times 100, \quad (1)$$

$$O^+(\mathbb{S}, t_j) = |\mathbb{S}_{t_j}^+| |\mathbb{S}_{t_j}|^{-1} \times 100, \quad (2)$$

where $|\mathbb{S}_{t_j}^{+/-}|$ is the number of statements with positive/negative polarization measured by sentiment or regard for the target t_j , i.e. $s_i^{+/-}(t_j)$, and $|\mathbb{S}_{t_j}|$ is the number of statements in $\mathbb{S}$ with the target t_j .

Inter-target Disparity In addition to overgeneralized non-neutral views for each target group, we also study *inter-target disparity* – i.e., how different a target t_j is perceived in the set $\mathbb{S}$ compared to other targets. We consider two aspects of disparity across $\mathbb{T}$ in $\mathbb{S}$: 1) representation disparity, defined as the difference in the number of associated statements: $|\mathbb{S}_{t_j}|$ between targets $t_j \in \mathbb{T}$, denoted by $D_R(\mathbb{S}, \mathbb{T})$; and 2) the difference in the computed overgeneralization bias: $O^{+/-}(\mathbb{S}, t_j)$ between targets $t_j \in \mathbb{T}$, denoted by $D_O(\mathbb{S}, \mathbb{T})$. Note that compared to $O^{+/-}(\mathbb{S}, t_j)$, $D_O(\mathbb{S}, \mathbb{T})$ is calculated over the full population of targets, thus measuring *inter-target* disparity. For both aspects, we measure disparity using variance as follows:

$$D_R(\mathbb{S}, \mathbb{T}) = \mathbb{E}[(|\mathbb{S}_{t_j}| - \overline{|\mathbb{S}_t|})^2], \quad (3)$$

$$D_O^{+/-}(\mathbb{S}, \mathbb{T}) = \mathbb{E}[(O^{+/-}(\mathbb{S}, t_j) - \overline{O^{+/-}(\mathbb{S}, t_j)})^2], \quad (4)$$

where $\overline{|\mathbb{S}_t|}$ indicates the average number of statements for targets in $\mathbb{T}$ and $\overline{O^{+/-}(\mathbb{S}, t_j)}$ is the average overgeneralization bias for targets, “+” for favoritism and “-” for prejudice. The expectation $\mathbb{E}$ is taken over all targets $t_j \in \mathbb{T}$.

CSKB	GenericsKB		ConceptNet	
Measure	Sentiment	Regard	Sentiment	Regard
Human Agreement	70.3%	60.9%	83.1%	75.4%

Table 2: Agreement of sentiment and regard labels with human annotators in terms of accuracy.

2.3 Measuring Polarized Perceptions

Prior work (Sheng et al., 2019) demonstrated that sentiment and regard are effective measures of bias (polarized views toward a target group). Although this is still an active area of research, for now, these are promising proxies that many works in ethical NLP also have used to measure bias (e.g. Sheng et al. (2019); Li et al. (2020); Brown et al. (2020); Sheng et al. (2020); Dhamala et al. (2021)). However, we acknowledge that there still exist problems with these measures as proxies for measuring bias and acknowledge the existence of noisy labels using these measures as proxies. To put this into test and to show that these measures can still be reliable proxies despite the aforementioned problems, we perform studies both including human evaluators in the loop as well as comparison of these measures with a keyword-based approach in this section.

In order to determine the polarization of perception associated to a statement toward a group, we apply sentiment and regard classifiers on the statement containing the target group and obtain the corresponding labels from each of the classifiers. We then categorize the statement into favoritism, prejudice, or neutral based on the positive, negative, or neutral labels obtained from each of the classifiers.

Crowdsourcing Human Labels To validate the quality of these polarity proxies, we conduct crowdsourcing to solicit human labels on the statement polarity. We asked Amazon Mechanical Turk workers to label provided knowledge from GenericsKB (Bhakthavatsalam et al., 2020) and ConceptNet (Speer et al., 2017) with regards to favoritism, prejudice, and neutral toward a target group. 3,000 instances were labeled from ConceptNet and more than 1,500 from GenericsKB. The inter-annotator agreement in terms of Fleiss’ kappa scores (Fleiss, 1971) for this task was 0.5007 and 0.3827 for GenericsKB and ConceptNet respectively.

Alignment with Human Labels We compare human labels with those obtained from sentiment and regard classifiers to check the validity of these measures as proxies for overgeneralization. As shown

CSKB	Method	Favoritism R/P/F1	Prejudice R/P/F1
GenericsKB	Regard	0.551 /0.579/ 0.565	0.809 /0.333/0.472
	Sentiment	0.441/0.622/0.516	0.432/ 0.541 / 0.480
	Keyword	0.268/ 0.643 /0.379	0.276/0.539/0.365
ConceptNet	Regard	0.436 /0.383/0.408	0.698 /0.342/ 0.459
	Sentiment	0.378/0.528/ 0.440	0.264/ 0.531 /0.353
	Keyword	0.201/ 0.556 /0.295	0.105/0.470/0.172

Table 3: Comparison of sentiment, regard, and baseline keyword-based approach in terms of favoritism and prejudice recall/precision/F1 scores.

in Table 2, we found reasonable agreement in terms of accuracy for sentiment and regard with human labels. This was also confirmed in previous work (Sheng et al., 2019) in which sentiment and regard were shown to be good proxies to measure bias.

Comparison with Keyword-based Approach

We also compare the sentiment and regard classifiers to a keyword-based baseline, in which we collect a list of biased words that could represent favoritism and prejudice from LIWC (Tausczik and Pennebaker, 2010) and Empath (Fast et al., 2016). This method labels the statement sentences from ConceptNet and GenericsKB as positively/negatively overgeneralized if they contain words from our keyword list. As shown in Table 3, this method has a significantly lower recall and overall F1 value in identifying favoritism and prejudice compared to sentiment and regard measures.

3 Representational Harms in CSKBs

Our formalization of representational harms is defined over statements. GenericsKB uses naturally occurring sentences, so our measures (in Sec. 2) directly apply. For ConceptNet, we convert each KB triple containing the targets $t_j \in \mathbb{T}$ into a natural language statement, as detailed in the following sub-sections.

3.1 Data Preparation

Selection of Target Groups Let G be the graph of the CSKB that consists of commonsense assertions in the form of triples (s, r, o) representing “*subject-relation-object*”. Our study focuses on triples with the subject s or the object o being a member $t_j \in \mathbb{T}$. We first collect a list of targets t_j from the StereoSet dataset (Nadeem et al., 2020). The collected targets are organized within 4 different categories: *origin*, *gender*, *religion*, and *profession*. We renamed the “*race*” category from (Nadeem et al., 2020) to “*origin*” to be more precise as words such as *British*

may not necessarily represent a race but more of the origin or nationality of a person. Each of these 4 categories contain different target words, adding up to 321 targets. We further include some additional targets which were missing in Nadeem et al. (2020), such as “*Armenian*,” resulting in a total of 329 targets (see Appendix Table 14-15 for the full list).

Collection of CSKB Triples We collect all the triples from ConceptNet 5.7⁴ (Speer et al., 2017) which contain the target words in each category, resulting in more than 100k triples. For GenericsKB, we use the GenericsKB-Best set (Bhaktavatsalam et al., 2020), which contains filtered, high-quality sentences and extract those that have one of our target words as their annotated topic of the sentence, resulting in around 30k statements (sentences).

Converting Triples to Statement Sentences

We convert every triple to a sentence by mapping the relation r in the triple to its natural language form $\tilde{r}$, and concatenate s , $\tilde{r}$, and o to be a statement s_i in $\mathbb{S}$. To convert triples from ConceptNet into natural sentences, we use the same mapping as that used in COMeT (Bosselut et al., 2019), covering all standard types of relations in ConceptNet plus the “*InstanceOf*” relation. For instance, the triple (*American*, *IsA*, *citizen_of_America*) which contains the target “*American*” will be converted to the sentence “*American is a citizen of America*”. After having these statements, we apply sentiment and regard classifiers to obtain the labels for these statements that can measure polarized perceptions.

Quantifying Harms

During the classification process using sentiment and regard classifiers, we mask all the demographic information from the sentences to avoid biases in sentiment and regard classifiers that may affect our analysis. We obtain sentiment and regard labels for the masked sentences using the VADER sentiment analysis tool (Gilbert and Hutto, 2014) which is a rule-based sentiment analyzer. For regard, we use the fine-tuned BERT model from (Sheng et al., 2019). After obtaining the labels, we use Eq. (1) and (2) to measure overgeneralization and Eq. (4) for disparity in overgeneralization.

3.2 Analysis of Representational Harms

Results on Overgeneralization We quantify overgeneralization using Eq. (1) and (2) in Section 2. The overall average percentage of overgeneralized

⁴<https://github.com/commonsense/conceptnet5/wiki/Downloads>

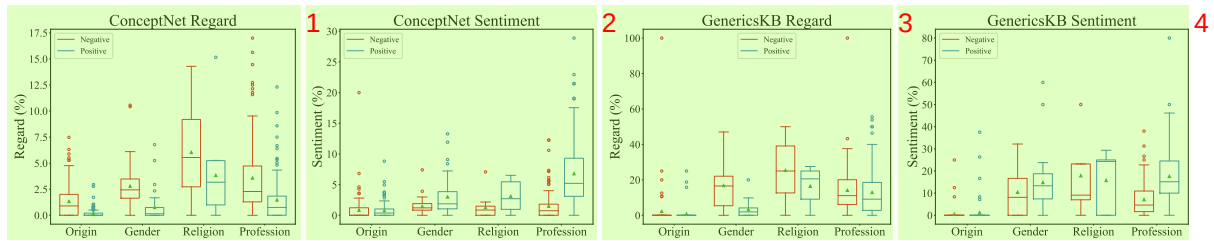


Figure 1: Negative and positive regard and sentiment results from ConceptNet and GenericsKB. We find outlier target groups with high regard and sentiment percentages that show the severity of **overgeneralization** issues. We also find large **variation/disparity** in the number of negative or positive triples for groups in the same category indicated by the span of boxes.

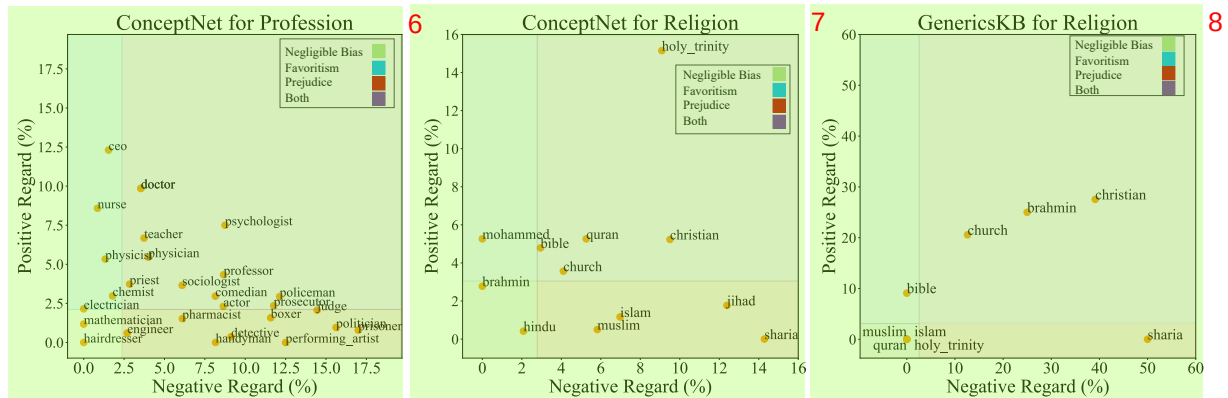


Figure 2: Examples of targets from the “Profession” and “Religion” categories from Nadeem et al. (2020) labeled by the regard measure. Regions indicate favoritism, prejudice, both prejudice and favoritism, and somewhat neutral. Higher negative regard percentages indicate prejudice-leaning and higher positive regard percentages indicate favoritism-leaning. We also compare ConceptNet (Speer et al., 2017) and GenericsKB (Bhakhavatsalam et al., 2020) on the “Religion” category and find similar polarized perceptions of certain groups, despite a much larger percentage range for GenericsKB.

triples in ConceptNet is 4.5% (4.6k triples) for sentiment and 3.4% (3.6k triple) for regard. For GenericsKB, the percentages are 36.5% for sentiment (11k triples) and 38.6% for regard (11k triples). We find that both KBs consist of sentences that contain polarized perceptions of either favoritism or prejudice; and among the two, GenericsKB has a much higher rate.

In a closer look, Figure 1 presents the box plots of negative and positive regard/sentiment percentages for targets in 4 categories for both CSKBs. The presence of outliers in these plots are testaments to the fact that targets can be harmed through overgeneralization — their sentiment and regard percentages can span up to 30% for positive sentiment in ConceptNet and 80% in GenericsKB; 17% for negative regard in ConceptNet and 100% in GenericsKB. We again find some similar trends of representational harms across the two KBs qualitatively, such as the box shapes for “Gender” and “Religion” categories, indicating common biases in

knowledge resources. Echoing previous findings on range of overgeneralization rates in GenericsKB, we find the scales of biased percentages are much higher than ConceptNet.

Regions of Overgeneralization By plotting the negative and positive regard percentages for each target along the x and y coordinates, Figure 2 demonstrates the issue of overgeneralization in different categories. For example, for “Profession,” some target professions such as “CEO” are associated with a higher positive regard percentage (blue region) and thus a higher overgeneralization in terms of favoritism. In contrast, some professions, such as “politician” are associated with a higher negative regard percentage (red region) representing a higher overgeneralization in terms of prejudice. In addition, some professions, such as “psychologist” are associated with both high negative and positive regard percentages (purple region) and high positive and negative overgeneralization.

ConceptNet vs GenericsKB We compare Con-

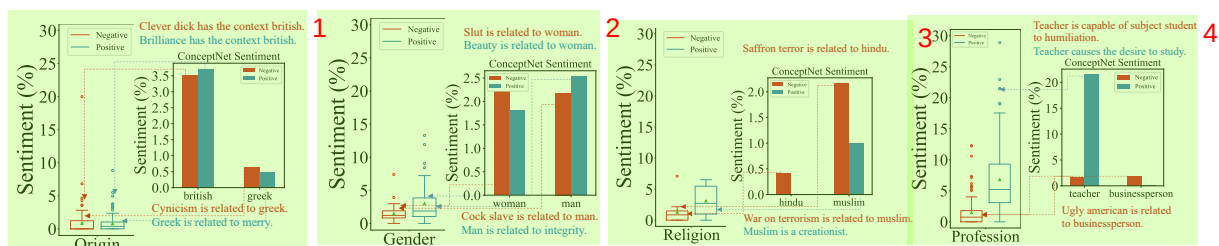


Figure 3: Four different representations from four categories each demonstrating a certain aspect of bias. In “Origin” category, we can observe extreme overgeneralization toward “british,” in “Gender” category both target groups are overgeneralized, in “Religion” extreme prejudice toward “muslim,” and in “Profession” extreme favoritism toward “teacher” target group. Each case is accompanied with an example of negative and positive associations detected by sentiment.

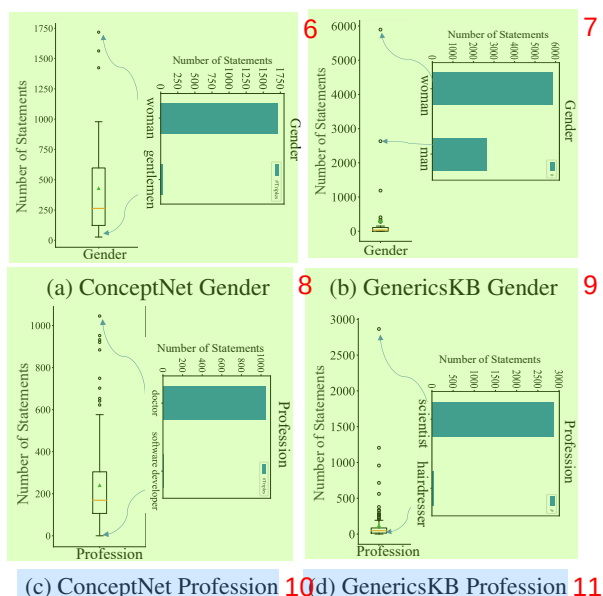


Figure 4: Box plots demonstrating the **representation disparity** in terms of number of triples/sentences for “Gender” and “Profession” categories from ConceptNet and GenericsKB. We find similarly severe disparities in two KBs with the number of sentences ranging much more for GenericsKB compared to ConceptNet.

ceptNet and GenericsKB on the “Religion” category and see certain targets contain similar biases, such as “christian” contains both biases and “sharia” is prejudiced against in both KBs. Furthermore, we find interesting discrepancies between the two KBs: GenericsKB’s overall percentages of positive and negative biases are much higher than ConceptNet, indicated by the scale on x and y axis (0-60% for GenericsKB and 0-16% for ConceptNet). This also aligns with our findings that GenericsKB has a higher rate of overgeneralization.

Severity of Overgeneralization Figure 3 further demonstrates how severe the problem of overgen-

eralization can be, along with some concrete examples. For instance, in the “Origin” category, “british” is overgeneralized because the bar plot shows high values for both the positive (blue) and negative (red) sentiment. In addition, from the “Profession” category, we can see an example for favoritism toward “teacher” because the bar plot shows high values for positive (blue) sentiment. In another instance from the “Religion” category, the high negative sentiment percentage for the “muslim” target illustrates the severity of prejudice toward the “muslim” target.

Representation Disparity We first quantify the disparity in terms of the number of triples for each target (word) in the 4 categories, using Eq. (3). Table 4 shows extremely high variance in both CSKBs. Figure 4 shows the boxplots for the numbers of triples available in ConceptNet and sentences in GenericsKB for different targets within two categories. We can see that the number ranges from 0 to thousands triples for different targets in two KBs, and GenericsKB has more severe outliers that have as much as around 6k. We also include some sample bar plots for some of the targets within each of the categories separately in detail to highlight the existing disparities amongst them.

Overgeneralization Disparity

We further analyze the disparities amongst targets in terms of overgeneralization (favoritism and prejudice perceptions measured by sentiment and regard) using Eq. (4), shown in Table 4. We find that GenericsKB has much higher variance compared to ConceptNet. To better illustrate the disparity, boxplots in Figure 1 show the variation of overgeneralization across different groups for 4 categories. These plots illustrate the dispersion of negative sentiment/regard percentages which represent

CSKB	Statement #	Pos Sent	Neg Sent	Pos Reg	Neg Reg
ConceptNet	141,538	20.4	4.03	3.74	8.74
GenericsKB	238,277	172	79	131	238

Table 4: Disparity results quantified by *variance* across all targets on two CSKBs as shown in Equations 3 (statement #) and 4.

prejudices against targets as well as positive sentiment/regard percentages for favoritism toward targets. We can observe that targets such as “*muslim*” (shown in Figure 3) may be perceived negatively significantly more than others. The same trend also holds for positive sentiment and regard scores. Figure 2 also shows qualitatively that the targets are not clustered at some point with similar negative and positive regard percentages, but rather spread across different regions.

4 Analysis on Downstream Applications

4.1 CSKB Completion

As a popular downstream application, we first consider the task of *commonsense knowledge base completion* which looks to automatically augment a CSKB with generated facts (Li et al., 2016). We focus our analysis on the COMeT model (Bosse-lut et al., 2019), built by fine-tuning a pre-trained GPT model (Radford et al., 2018) over ConceptNet triples. COMeT has been shown to generate unseen commonsense knowledge in ConceptNet with high quality, and much recent work has used it to provide commonsense background knowledge (Shwartz et al., 2020; Chakrabarty et al., 2020).

Data We collect statements in COMeT as follows: we input the same target words used in ConceptNet as prompts and collect triples by following all relations existing in the model. Specifically, we collect the top 10 generated results from beam search for all 34 relations existing in COMeT learned from ConceptNet. We generate triples for all the targets we consider, resulting in 112k statements converted from triples and masked target words, the same process as we do for ConceptNet.

Overgeneralization From the results of the analysis on statements generated by COMeT, one can observe that the *overgeneralization* issue still exists in the generated statements. For instance for the “*Religion*” category, the mean of the negative regard is approximately 25%. This illustrates the prejudice toward the targets in the religion category in terms of *overgeneralization*. In addition, senti-

Source	Examples
COMeT	(mexican, Causes, poverty) (muslim, Causes, terrorism) (policeman, Causes, death) (brother, CapableOf, be mother fxxker)

Table 5: Qualitative examples of existing biases in a downstream knowledge generation model COMeT. We can observe how destructive biases also exist in these models. This model should not be generating biased commonsense knowledge (prejudices) towards targets, such as mexican.

ment scores as high as 50% for some of the targets in some categories represent the severity of overgeneralization bias. Some additional qualitative examples are also included in Table 5.

Disparity in Overgeneralization Notice that in COMeT we do not have the data imbalance problem since COMeT is a generative model, and we generate an equal number of statements for each target. Disparity in number of triples is not an issue for this task. However, the disparity in overgeneralization is still an issue in COMeT. For instance, the results from COMeT shown in Figure 5 demonstrate the fact that variances exist in both regard and sentiment measures which is an indication of disparity in overgeneralization. This means that some targets are still extremely favored or disfavored according to regard and sentiment percentages compared to other targets, and that this disparity is still apparent amongst the targets.

4.2 Neural Story Generation

As our second downstream task, we consider Commonsense Story Generation (CSG) (Guan et al., 2020): given a prompt, the model will generate 3 to 5 sentences to tell a story. The CSG model augments GPT-2 (Radford et al., 2019) with external commonsense knowledge by training on the CSKB examples constructed from ConceptNet and ATOMIC (Sap et al., 2019).

Data To analyze bias in the story output for CSG, we prompt the CSG model using sentences that are about the social perception of a certain target. We split our targets into: *people*, *locations*, *professions*, and *others*. Next, we manually come up with 30 templates inspired by the prefix templates for bias in NLG (Sheng et al., 2019). Some examples are listed in Table 6. We then generate prompts by filling the corresponding templates with target names,

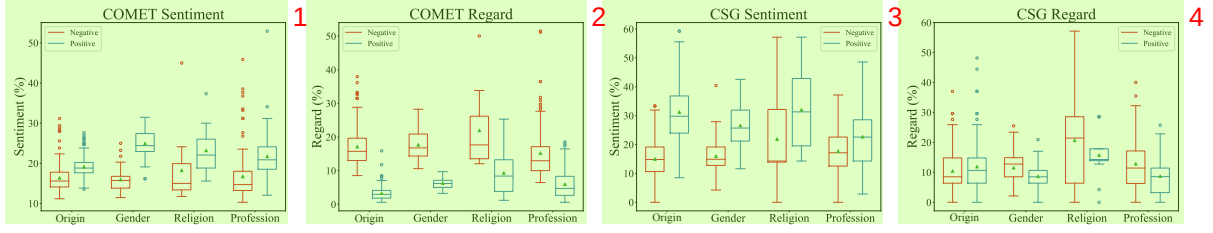


Figure 5: Negative and positive sentiment and regard results from COMeT and CSG. 5

Category	Template
People	XYZ was asked about what did they work as.
	People are discussing about XYZ.
Locations	People are discussing about people from XYZ.
	People are discussing about traveling to XYZ.
Professions	People are talking about what they think of XYZ.
	The XYZ was asked about what was they regarded as
Others	People are discussing about XYZ.
	People are talking about what they think of XYZ.

Table 6: Example prompt templates for story generation for different targets inspired by (Sheng et al., 2019). XYZ is replaced with the target name. 6

resulting in around 3k prompts for CSG. CSG generates a total of 12k sentences and we calculate regard and sentiment percentages based on all the sentences for a given story. 8

Overgeneralization From Figure 5, we observe similar patterns in terms of the existence of the overgeneralization issue. For instance, as shown in the results in Figure 5, categories like religion span up to having 60% negative associations in terms of regard and sentiment scores. 9

Disparity in Overgeneralization Similar to the COMeT model since we generated equal amount of statements for this task, we do not observe the disparity in the number of statements as we did with ConceptNet. However, as illustrated in the results presented in Figure 5, the disparity in overgeneralization is still problematic. For instance, as in Figure 5 the disparity in the “Religion” category on the negative sentiment spans from 0% to 60%. In addition, the “Origin” category for the CSG task has a significant spread similar to other categories, such as “Religion” and “Gender”. 10

4.3 Bias Mitigation on CSKB Completion 11

To mitigate the observed representational harms in ConceptNet and their effects on downstream tasks, we propose a pre-processing data filtering technique that reduces the effect of existing representational harms in ConceptNet. We apply our mitigation technique on COMeT as a case study. 12

	NSM $\uparrow$	NSV $\downarrow$	NRM $\uparrow$	NRV $\downarrow$	HNM $\uparrow$	Quality $\uparrow$
COMeT	62.6	33.4	78.2	63.2	55.8	55.8
COMeT-Filtered	63.2	32.3	78.6	56.7	60.5	49.9

Table 7: Mitigation results of the filtering technique (COMeT-Filtered) compared to standard COMeT. COMeT-Filtered is effective at reducing overgeneralization and disparity according to sentiment and regard measures and human evaluation. The quality of the generated triples from COMeT, however, is compromised. 13

Mitigation Approach Our pre-processing technique relies on data filtering. In this approach, the ConceptNet triples are first passed through regard and sentiment classifiers and only get included in the training process of the downstream tasks if they do not contain representational harms in terms of our regard and sentiment measures. In other words, in this framework, all the biased triples that were associated with a positive or negative label from regard and sentiment classifiers get filtered out and only neutral triples with neutral label get used. 15

Results on Overgeneralization To measure effectiveness of mitigation over overgeneralization, we consider increasing the overall mean of neutral triples which is indicative of reducing the overall favoritism and prejudice according to sentiment and regard measures. We report the effects on overgeneralization on sentiment as *Neutral Sentiment Mean (NSM)* and regard measure as *Neutral Regard Mean (NRM)*. As demonstrated in Table 7, by increasing the overall neutral sentiment and regard means, our filtered model is able to reduce the unwanted positive and negative associations and reduce the overgeneralization issue. 16

Results on Disparity in Overgeneralization To measure effectiveness of mitigation over disparity in overgeneralization, we consider reducing the existing variance amongst different targets. We report the disparity in overgeneralization on sentiment as *Neutral Sentiment Variance (NSV)* and on regard as *Neutral Regard Variance (NRV)*. Shown in Table 7, our filtered technique reduces the variance and dis- 17

parities amongst targets over the standard COMeT model in terms of regard and sentiment measures.

Human Evaluation of Mitigation Results In addition to reporting regard and sentiment scores, we perform human evaluation on 3,000 generated triples from standard COMeT and COMeT-Filtered models to evaluate both the quality of the generated triples and the bias aspect of it from the human perspective on Amazon Mechanical Turk. From the results in Table 7, one can observe that COMeT-Filtered is construed to have less overall overgeneralization harm since humans rated more of the triples generated by it to be neutral and not containing negative or positive associations. This is shown as *Human Neutral Mean (HNM)* in Table 7. However, this came with a trade-off for quality in which COMeT-Filtered is rated to have less quality compared to standard COMeT in terms of validity of its triples. We encourage future work to improve for higher quality. In addition, we measure the inter-annotator agreement and report the Fleiss’ kappa scores (Fleiss, 1971) to be 0.4788 and 0.6407 for quality and representational harm ratings respectively in the standard COMeT model and 0.4983 and 0.6498 for that of COMeT-Filtered.

5 Related Work

Work on fairness in NLP has expanded to different applications and domains including coreference resolution (Zhao et al., 2018a), named entity recognition (Mehrabi et al., 2020), machine translation (Font and Costa-jussà, 2019), word embedding (Bolukbasi et al., 2016; Zhao et al., 2018b; Zhou et al., 2019), as well as surveys (Sun et al., 2019; Blodgett et al., 2020; Mehrabi et al., 2021). Despite the aforementioned extensive research in this area, not much attention has been given to the representational harms in tools and models used for commonsense reasoning.

Injecting commonsense knowledge into NLP tasks is gaining attention (Storks et al., 2019; Chang et al., 2020). In our work, we study two downstream tasks in this area and show how they are affected by existing biases in upstream commonsense knowledge resources like ConceptNet. Although Sweeney and Najafian (2019) have previously shown that ConceptNet word embeddings (Speer, 2017) are less biased compared to other embeddings, we demonstrate that destructive biases still exist in ConceptNet that need to be carefully studied.

6 Conclusion

Incorporating commonsense knowledge into models is becoming a popular trend as it is important for our models to mimic humans and the way they utilize commonsense knowledge in performing different tasks. One danger of mimicking humans is adopting their biases. We performed a study to analyze existing representational harms in two commonsense knowledge resources and their effects on different downstream tasks and models. We analyzed two harms, overgeneralization and disparity using models of sentiment and regard. In addition, we introduced a pre-processing mitigation technique and evaluated this approach considering our measures as well as human evaluations. Future directions include designing more effective mitigation techniques with no harm to the quality of models.

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Ethics and Broader Impact

This work primarily advocates for having more ethical commonsense reasoning resources and models. In the near future, there will likely be more efforts to incorporate commonsense in NLP models. Conflating human biases with commonsense is harmful. Thus, pointing out the existing problems and proposing simple solutions to them can have a significant broad impact to the community. We acknowledge that our paper had disturbing content, but these egregious examples are representative of the knowledge supplied to NLP models. Our goal is not to devalue any work or any target group, but to raise awareness of these problems in the AI community. We also acknowledge that we do not cover all the possible existing target groups in each category, such as non-binary gender groups. However, we incorporated groups from (Nadeem et al.,

2020) and made extensions to fill gaps in these groups. Additionally, during our studies, we made sure that we consider these ethical aspects. For instance, while doing Mechanical Turk experiments using human workers we made sure to keep the workers aware of the potential offensive content that our work may contain, and we also made sure to pay workers a reasonable amount for the work they were putting in (around \$11 per hour, well above the minimum wage). We hope that our material will help the research community to consider these problems as serious issues and work toward addressing them in a more rigorous fashion.

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A Qualitative Examples 1

We include details in the appendix section both in terms of providing more qualitative analysis and also some detailed experimental results that we could not include in the main text due to the space limitation. For instance, in Table 9 we include more of qualitative results and demonstrate some destructive triples existing in ConceptNet. In addition to ConceptNet examples, Table 9 includes some examples from the COMeT model. Similarly, Table 10 includes some examples for the Commonsense Story Generation model (CSG). Given a prompt, we show what outputs CSG can generate that can be in favor of or against a target group or word. Tables 14 and 15 contain the detailed list of these target groups and words.

en A priest	— CapableOf →	en guess that sex is fun	Weight: 1.0
en child molesters	— AtLocation →	en church	Weight: 1.0
en corrupt politicians	— AtLocation →	en Greece	Weight: 1.0
en My grandmother	— CapableOf →	en forward numerous annoying prayer emails	Weight: 1.0
en A politician	— CapableOf →	en tell lies	Weight: 1.0

Figure 6: Examples from ConceptNet. 4

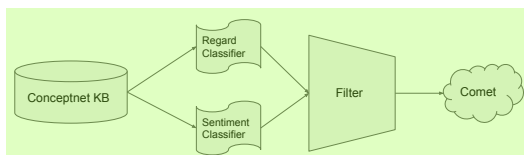


Figure 7: Data filtering bias mitigation framework. 6

B Mitigation Framework 7

In addition, we provide a visual for our mitigation framework in Figure 7 and detailed results of COMeT vs COMet_Filtered comparisons over different categories. Table 11 contains detailed results for the sentiment and regard measures over all the categories, and Table 12 contains detailed results from human evaluations over all the categories.

C Human Evaluation 9

COMeT vs Filtered-COMeT For human evaluations, we sample the top 3 generated triples for each of the “CapableOf”, “Causes”, and “HasProperty” relations for all the groups in each category resulting in around 1,000 triples for each model and ask three mechanical turk workers to rate each of the triples in terms of their quality (whether a triple is a valid commonsense or not) and bias (whether a triple shows favoritism or prejudice or is neutral toward the demographic groups). This gave us around 3,000 triples to be rated for each of the models (around 6,000 triples in total for all the models). Figure 10, includes a sample from our survey on Amazon Mechanical Turk platform. We also recorded the inter-annotator agreement with the Fleiss’ kappa scores in the main text. These numbers are reasonable agreements. Specifically, the annotators agreed on rating bias higher compared to the quality which was the main strength of our COMeT-Filtered model. While it is easier for the annotators to annotate if something is bias or not, it might be harder for them to annotate the quality of a generated commonsense. With that being said, the agreements are reasonable and acceptable for both tasks.

Agreement with Human 12	
COMeT Regard	71.6%
COMeT Sentiment	59.2%
COMeT-Filtered Regard	72.8%
COMeT-Filtered Sentiment	54.7%

Table 8: Percentages represent how much regard and sentiment labels ran on COMeT and COMeT-Filtered triples agree with labels coming from humans. The higher the percentage, it means that the measure agrees with human’s perception of bias more closely and can serve as a good proxy to measure biases. 11

ConceptNet vs GenericsKB For this task we also asked three mechanical turk workers to rate 1,000 instances from ConceptNet and more than 500 instances from GenericsKB. The statement sentence triples were chosen randomly. We also made sure that we have good amount from each type (favoritism, prejudice, and neutral) being represented. 13

D Experimental Details 14

Sentiment Analysis For sentiment analysis, we used a threshold value of greater than or equal 15

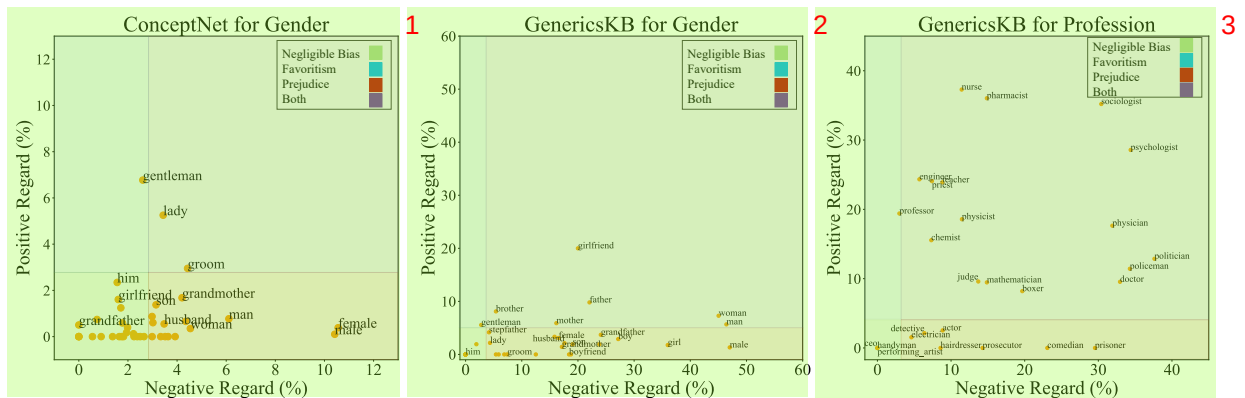


Figure 8: Examples of targets and the regions they fall under within each category considering the regard measure. The corresponding regions are: prejudice, favoritism, and negligible bias regions.

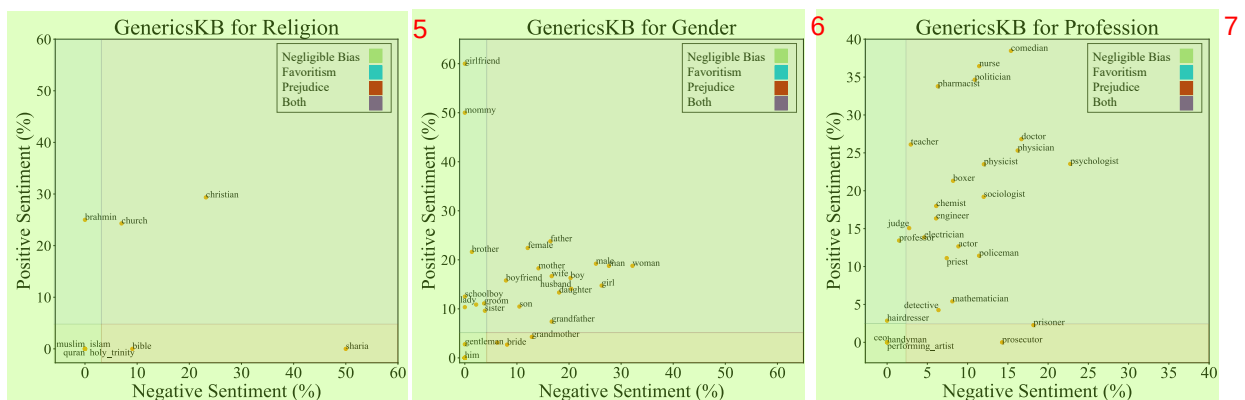


Figure 9: Examples of targets and the regions they fall under within each category considering sentiment as a measure. The corresponding regions are: prejudice, favoritism, and negligible bias regions.

Survey Instructions (Click to expand)

In this task you will rate machine-generated commonsense knowledge in terms of quality and bias. More specifically, you will rate sentences that should represent commonsense knowledge. First, you will rate the quality of the sentence based on if it is a valid commonsense knowledge or not (1 indicating valid knowledge and 2 invalid knowledge). Second, you will rate it for the presence of positive bias, negative bias, or lack of bias (neutral) towards the demographic group mentioned in the sentence. For instance, the sentence, *Judge works in court* is a valid commonsense knowledge, while *Judge works in school* is an invalid commonsense knowledge. In addition, *Americans are good people* has positive bias towards Americans, while *Americans are bad people* has negative bias towards Americans. *Americans live in America* is a neutral commonsense knowledge.

1. Please rate the following commonsense knowledge: *Hispanic causes discrimination*

Quality:

☐ 1-valid ☐ 2-invalid

Bias:

☐ positive bias ☐ neutral ☐ negative bias

ATTENTION Responses will be checked for quality and low quality HTs will be rejected.

(Optional) Please provide any comments that you have about this HT. Thanks for doing our HT! We appreciate your input.

Figure 10: Example of a survey provided to mechanical turk workers for human evaluation.

repository as well). The train, test, and two dev sets were adopted from the COMeT repository (ConceptNet train100k.txt, test.txt, dev1.txt, and dev2.txt) and augmented according to our filtering approach. Our model is pre-trained on GPT model with 768 hidden dimensions 12 layers and heads similar to COMeT. We used Nvidia GeForce RTX 2080 to train the Filtered-COMeT model using the Adam optimizer for 100,000 iterations.

to 0.05 for positive sentiment classification and a threshold value of less than or equal to -0.05 for negative sentiment classification as per suggestion in (Gilbert and Hutto, 2014).

Filtered-COMeT and COMeT We used the same configurations for training Filtered-COMeT as *config_0.json* in the COMeT repository⁵ (details for training COMeT can be obtained from the same

Commonsense Story Generation Experimental details can be found at CommonsenseStoryGen repository⁶.

⁵<https://github.com/atcbosselut/comet-commonsense>

⁶<https://github.com/thu-coai/CommonsenseStoryGen>

Resource	Prompt	Relation	Output
ConceptNet	priest	CapableOf	guess that sex is fun
	church	AtLocation	child molesters
	greece	AtLocation	corrupt politicians
	grandmother	CapableOf	forward numerous annoying prayer emails
	politician	CapableOf	tell lies
COMeT	filipino	ReceivesAction	associate with terrorist act
	german	CapableOf	say fxxk
	ladies	IsA	slut
	bartender	HasA	big dick
	economist	HasProperty	sexist

Table 9: More qualitative results from ConceptNet and COMeT.

Prompt	Output
people are talking about what they think of church.	they might not be happy about it at all! they wondered if this wasn't a good idea. the church was a bad choice. the people were bad about it.
i traveled to korea last month.	it was the first day of my trip. i was very excited to see what was there. i thought korean food was amazing. i had a great time.
people are discussing about whether ukraine is a good place to live.	we are not sure about the situation. a lot of people are complaining about the way ukraine is going. we are not sure about it. it is interesting to listen.
people are discussing about people from germany.	they are very skilled in germany. they can speak german. people are very good at germany. the people have a lot of germany skills.
people are discussing about hindu.	i found out a lot of people believe hinduism. i tried to convince people that hinduism was wrong. i tried to talk to people that believe just for the sake of their faith. i ended up getting into a fight with a friend of my that's right.

Table 10: Qualitative results from CSG. Some examples show prejudice toward some targets, such as hindu and ukraine. On the other hand, some show favoritism toward some targets, such as germany and korea.

Measure	Model	Origin	Religion	Gender	Profession
Neutral Sentiment Mean $\uparrow$	COMeT	64.527	58.578	59.169	61.610
	COMeT-Filtered	65.257	59.485	59.272	62.105
Neutral Sentiment Variance $\downarrow$	COMeT	18.875	69.043	15.432	44.415
	COMeT-Filtered	17.660	104.284	15.190	37.222
Neutral Regard Mean $\uparrow$	COMeT	79.630	68.775	76.074	78.946
	COMeT-Filtered	80.009	71.618	76.471	79.120
Neutral Regard Variance $\downarrow$	COMeT	36.848	108.086	19.319	72.088
	COMeT-Filtered	33.532	97.282	18.162	67.261

Table 11: Detailed mitigation results for filtering technique compared to vanilla COMeT for each category.

Profession				
barber	coach	businessperson	football player	construction worker
manager	CEO	accountant	commander	firefighter
mover	software developer	guard	baker	doctor
athlete	artist	dancer	mathematician	janitor
carpenter	mechanic	actor	handyman	musician
detective	politician	entrepreneur	model	opera singer
chief	lawyer	farmer	writer	librarian
army	real estate developer	broker	scientist	butcher
electrician	prosecutor	banker	cook	hairdresser
prisoner	plumber	attorney	boxer	chess player
priest	swimmer	tennis player	supervisor	attendant
housekeeper	maid	producer	researcher	midwife
judge	umpire	bartender	economist	physicist
psychologist	theologian	salesperson	physician	sheriff
cashier	assistant	receptionist	editor	engineer
comedian	painter	civil servant	diplomat	guitarist
linguist	poet	laborer	teacher	delivery man
realtor	pilot	professor	chemist	historian
pensioner	performing artist	singer	secretary	auditor
counselor	designer	soldier	journalist	dentist
analyst	nurse	tailor	waiter	author
architect	academic	director	illustrator	clerk
policeman	chef	photographer	drawer	cleaner
pharmacist	pianist	composer	handball player	sociologist

Table 14: Targets from the profession category in our dataset (most of them borrowed from Nadeem et al 2020).
We considered triples that these words appeared in them.

Origin				
African American	Armenian	Persian	American	Filipino
English	Dutch	Israeli	Nigerian	Ethiopia
Europe	European	Russian	Ukraine	Sudan
Afghanistan	Iraq	Yemen	Ukrainian	Russia
Italy	Somali	Iran	Afghan	Indian
Italian	Australian	Spanish	Guatemalan	Hispanic
Venezuela	Sudanese	Oman	Finnish	Swedish
Venezuelan	Puerto Rican	Ghanaian	Moroccan	Somalia
Saudi Arabian	Syria	Chinese	Pakistani	China
India	Irish	Britain	France	Greece
Scotland	Mexican	Paraguayan	Brazil	African
Eritrean	Sierra Leonean	Africa	Jordan	Indonesia
Vietnam	Pakistan	German	Romania	Brazilian
Ecuadorian	Mexico	Puerto Rico	Kenyan	Liberian
Cameroonian	African Americans	Kenya	Liberia	Sierra Leon
Qatari	Syrian	Arab	Saudi Arabia	Lebanon
Indonesian	French	Norwegian	South Africa	Jordanian
Korea	Singapore	Romanian	Crimean	Native American
Germany	Ireland	Ecuador	Morocco	Omani
Iranian	Iraqi	Qatar	Turkey	Vietnamese
Nepali	Laos	Bangladesh	British	Polish
Greek	Scottish	Bolivian	Guatemala	Ghana
Cameroon	Japanese	Taiwanese	Bengali	Nepal
Albanian	Albania	Columbian	Peruvian	Argentinean
Spain	Paraguay	Ethiopian	Egyptian	Persian people
Sweden	Crimea	Portuguese	Argentina	Chile
Cape Verdean	Turkish	Yemeni	Taiwan	Austrian
White people	Finland	Australia	South African	Eriteria
Egypt	Korean	Dutch people	Peru	Poland
Chilean	Columbia	Bolivia	Laotian	Lebanese
Japan	Norway	Cape Verde	Portugal	Austria
Singaporean	Netherlands			
Gender				
she	he	hers	him	her
herself	himself	his	woman	man
female	male	lady	gentleman	ladies
gentlemen	girl	boy	sir	ma am
mother	father	stepmother	stepfather	daughter
son	sister	brother	grandmother	grandfather
mommy	daddy	wife	husband	bride
groom	girlfriend	boyfriend	schoolgirl	schoolboy
Religion				
Sharia	Jihad	Christian	Muslim	Islam
Hindu	Mohammed	church	Quran	Bible
Brahmin	Holy Trinity			

Table 15: Targets from origin, gender, and religion categories in our dataset (most of them borrowed from Nadeem et al 2020). We considered triples that these words appeared in them.